

In-Class Problem Set #15: September 23, 2026

1a. Alice rolls a 6-sided dice, and Bob rolls an 8-sided dice. Let X denote the sum of the two values. Is X a Binomial random variable? Why or why not?

1b. Suppose a drawer of t-shirts contains 7 t-shirts from Purdue and 3 t-shirts from IU. Alice randomly selects t-shirts from the drawer, **without replacement**, until the first IU t-shirt. All possible orders of selection are equally likely. Let X denote the number of t-shirts chosen, until the first IU t-shirt. Is X a Binomial random variable? Why or why not?

1c. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy; the weather types are independent and equally likely. Let X denote the number of sunny days during a 5-day weather prediction. Is X a Binomial random variable? Why or why not?

2a. Consider two independent Binomial random variables, X and Y , each with $n = 3$ and $p = 2/3$. Find the probability that neither of these random variables is 0, i.e., find $P(X \neq 0 \text{ \& } Y \neq 0)$.

2b. Consider m independent Binomial random variables, $X_1, \dots, X_m$, each with $n = 3$ and $p = 2/3$. Find the probability that none of these m variables are 0, i.e., find $P(X_1 \neq 0 \text{ \& } X_2 \neq 0 \text{ \& } \dots \text{ \& } X_m \neq 0)$.

3. Consider a random bear family of 3 bears: a daddy bear, a momma bear, and a baby bear. Each bear has six equally likely colors, one of which is green. The colors of the bears are independent. Let X denote the number of green bears in such a family.

3a. Is X a Binomial random variable? Why or why not?

3b. Let Y be the number of yellow bears in that very same (randomly selected) bear family as in **3a**. Is $X + Y$ a Binomial random variable? Why or why not? (Hint: $X + Y$ is the number of bears in the bear family that are either green or yellow.)

4. Consider the X and Y from Question **3a** and **3b**. Define a new random variable $W = X - Y$. Find the probability mass function of W .

5. Consider the X and Y from Question **3a** and **3b**.

5a. Find $P(X + Y > 0)$.

5b. Find $P(X - Y > 0)$.